

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages:** Python, SQL, Java(leetcode), C++(leetcode)

• **Data Engineering:** ETL/ELT Pipelines, Data Modeling, Snowflake, PySpark, Workflow Orchestration (Airflow, Step Functions, AWS Glue)

• **Cloud & Tools:** AWS (S3, EC2, EKS), PostgreSQL, Neo4j, Git, CI/CD, Docker, Apache Spark, Hadoop, Hive, NoSQL, Linux

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Built and maintained AWS-based scalable data pipelines using Apache Spark and distributed computing to ingest high-volume data for 100+ users, enabling S3 storage and downstream analytics
- Designed and optimized **SQL-based data models and warehouse tables** in PostgreSQL, applying indexing and query tuning techniques to improve performance by 38% for large datasets
- Implemented workflow orchestration using Airflow and AWS Step Functions for streaming pipelines, adding data quality checks, logging, retries, and failure handling to ensure reliability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed **scalable ELT pipelines** in **Python** for high-volume social media datasets in distributed environments, reducing processing latency by 41%
- Built distributed real-time ingestion pipelines using Kafka and Neo4j NoSQL graph database, enabling streaming data processing, transformation, and graph-based analytics at scale
- Applied Hadoop-based distributed processing and big data principles to monitor data quality, integrity, and system performance across large-scale streaming environments

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable **data processing pipelines** using **FastAPI**, **PostgreSQL**, and optimized storage layers to support large-scale analytics and downstream applications
- Designed fault-tolerant **ETL workflows** for ingesting and transforming high-volume document datasets, ensuring low-latency and reliable data
- Developed backend services and used **Git-based CI/CD workflows** to enable reliable deployment and real-time querying across distributed and high-availability environments

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built **data pipelines** feeding **Power BI dashboards** for logistics KPIs, reducing reporting time by **40%** and improving data availability for business decision-making
- Processed large-scale datasets using Python and Apache Spark distributed processing, implementing transformations and aggregations for system performance monitoring
- Developed **SQL queries** to validate, reconcile, and analyze operational data, improving data quality and speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Built and maintained **SQL-based data pipelines** supporting **Power BI dashboards**, designing data warehouse models, writing complex joins and aggregations, and ensuring delivery of analytics-ready datasets across business functions
- Automated **data transformation workflows** using VBA, implementing validation checks, error handling, and scheduling logic to streamline large-scale data processing and cut reporting time by **42%**

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Documented data workflows, pipeline architecture, and technical solutions to support cross-functional collaboration across engineering and analytics teams
- Managed **Agile workflows** using JIRA, tracking delivery, QA issues, and sprint progress to ensure timely and reliable releases

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

- Built an **end-to-end data pipeline** using Python, Pandas, and LightGBM to process large-scale **time-series** transaction data, including ingestion, transformation, feature engineering, and model integration
- Designed and implemented **scalable ETL and feature engineering pipelines** for large datasets, capturing behavioral trends and temporal patterns to improve model performance
- Developed a **data-driven decision framework**, incorporating pipeline outputs and risk thresholds to improve prediction precision by 20% and enable scalable analytics workflows